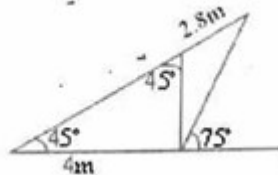


Section-A (25 Marks)

1. A hill road has a section as shown in the figure. Calculate the quantity of earthwork in a length of one kilometer. [10]



Answer:

Height of cutting:

$$h = 4 \tan 45^\circ = 4 \times 1 = 4 \text{ m}$$

Horizontal projection:

$$x = \frac{h}{\tan 75^\circ}$$

Since,

$$\tan 75^\circ = 3.732$$

$$x = \frac{4}{3.732} = 1.072 \text{ m}$$

The section consists of two triangles.

$$A_1 = \frac{1}{2} \times 4 \times 4 = 8 \text{ m}^2$$

$$A_2 = \frac{1}{2} \times 1.072 \times 4 = 2.144 \text{ m}^2$$

Therefore,

$$A = A_1 + A_2 = 8 + 2.144 = 10.144 \text{ m}^2$$

Quantity of earthwork for 1 km:

$$L = 1000 \text{ m}$$

$$V = 10.144 \times 1000 = 10144 \text{ m}^3$$

$$\text{Quantity of earthwork for 1 km} \approx 1.01 \times 10^4 \text{ m}^3$$

2. a. Describe with sketches the elements of highway cross sections.
b. Why are transition curve provided in the horizontal alignment of a highway explain with sketch. [5+5]

Answer:

a. Highway cross-section is the section of road cut perpendicular to the centerline axis of the road. The elements of highway cross-section influence the life of the pavement as well as riding comfort and safety. It consists of many parts such as carriageway, shoulder, median strips, traffic separators, side drain, side slope, catch drains etc.

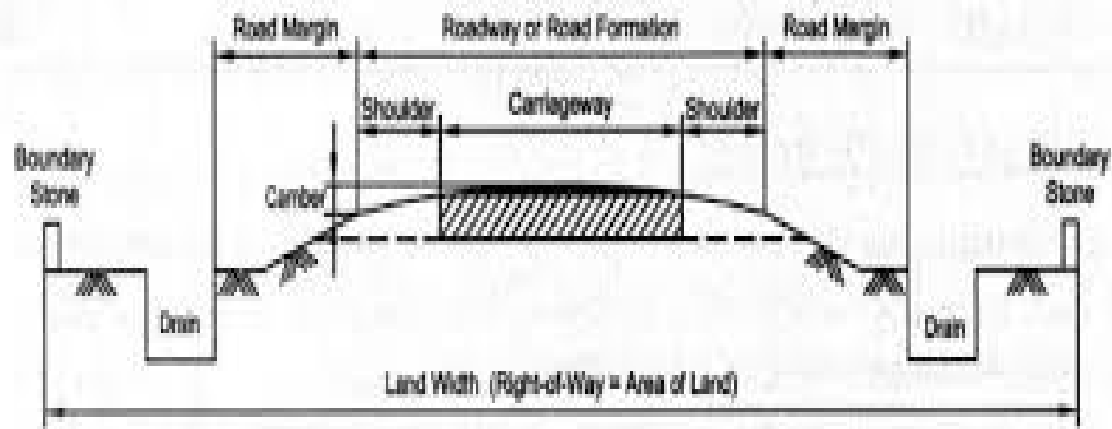


FIG. 1 CROSS-SECTION DETAILS OF A ROAD

i. Carriageway:

- It is the strip of road meant for vehicular traffic movement also known as pavement width.
- Width of carriageway depends upon the width of traffic lanes and number of lanes.
- According to NRS 2070, The minimum lane width for a single lane road is 3.75 m.
- Two land road is provided with minimum of 3.5 m for each lane.

ii. Shoulder:

- They are the strips provided along the road edge and are intended for the accumulation of stopped vehicles.
- Shoulders serve as emergency lane for vehicles and also provide the lateral support for base and surface courses.
- According to NRS 2070, the width of shoulders on either side of carriageway shall be at least 0.75m.
- It is desirable that color and texture of shoulders be different from those of the carriageway.

iii. Roadway width:

The sum of width of pavements or carriageway including separators and shoulders is called width of formation or roadway width.

It is the top width of highway embankment or bottom width of highway cutting excluding the side drains.

iv. Side slope:

- According to NRS 2070, side slopes of embankment and cuttings depend upon the type of fill/ cut materials and height/depth of fill or cut.
- If natural cross slope of the ground is more than 1:5 then the ground should be cut with more than 2 m wide horizontal steps.

v. Lay-bys, bus bays and bus stops:

- Lay-bys are the intermittent shoulders sufficiently wide and long provided to meet the important function of shoulder where the continuous shoulder on either side cannot be provided from economic considerations.
- Bus-bays shouldn't be located too close near the intersections. It is desirable that they are located at least 40 m from the intersection on either side.

- Bus stops should be placed adjacent to the bus linear line of travel so that it doesn't need to pull over to left. It must be provided at the spacings of 200-400 m.

vi. Right of way:

It is the width of land to be acquired for the road along its alignment. It must be adequate to accommodate all the cross-sectional elements of highway and reasonably provided for future development.

According to NRS2070,

ROW requirements for national highways= 50 m (25 m on either side of the road centerline)

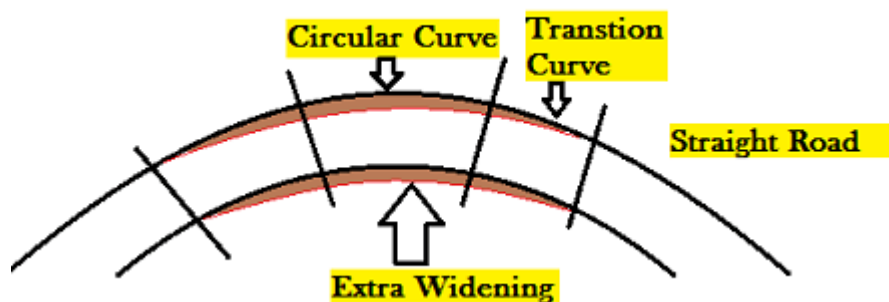
Feeder roads = 30 m (15 m on either side of road centerline)

District roads= 20 m (10 m on either side of road centerline)

b. Transition curve is provided to change the horizontal alignment from straight to circular curve gradually and has a radius which decreases from infinity at the straight end to the desired radius of circular curve at the other end.

Development of centrifugal force at horizontal curves causes discomfort to the passengers and also makes the steering of the vehicle difficult. Hence, transition curve is provided to keep the rate of introduction of centrifugal force consistent throughout the curve. It is provided in horizontal alignment of a highway:

- To introduce the centrifugal force gradually between the tangent point and beginning of the circular curve, avoiding sudden jerk on the vehicle.
- To enable the driver to turn the vehicle steering gradually for the comfort and security.
- To introduce gradually designed super elevation.
- To introduce gradually designed extra widening.
- To enhance the aesthetic appearance of the road.



3. a. What are the criteria that should be fulfilled while designing subsurface drains? [5]

b. What are the various tests that are used for assessing the suitability of road aggregates. [5]

Answer:

a. Highway drainage can be defined as the process of interception and removal of water from over, under and vicinity of road surface. It can be surface drainage or subsurface drainage. Removal and diversion of surface water from highway and adjoining land is called surface drainage. Similarly, removal of water from below the ground surface is called subsurface drainage.

Variation in moisture content of subgrade is caused by seepage of water, penetration of moisture through the pavement, rise and fall of ground water table and many more

The following criteria should be fulfilled while designing subsurface drains:

- The drainage system should maintain the groundwater table at the depth of 1-1.5 m from the ground surface.
- The subsurface drainage system should have sufficient capacity to remove excess water within the desired time after rainfall or irrigation.
- Drains should be kept at the depth of 1-2m to lower the water table effectively but not so deep such that the installation becomes uneconomical.

- iv. The design should achieve the required drainage performance at the minimum life-cycle cost while ensuring reliability.
- v. Drain spacing should be selected based on: Soil permeability (hydraulic conductivity), Drain depth, required water table level and rate of groundwater recharge.
- vi. Drain pipes should have adequate diameter to carry the expected discharge without flowing under pressure.
- vii. Adequate slope should be provided to maintain self-cleansing velocity and prevent sediment deposition.
- viii. The drainage system should have a safe and reliable outlet.
- ix. The outlet should remain above the receiving water level to prevent backflow and should be protected against erosion.
- x. Drain pipes should be corrosion-resistant and durable.

One of the commendable examples of good highway drainage in Nepal **is** the drainage arrangement along BP highway (Sindhuli road), where combination of cross drainage, side slope drainage and sub surface horizontal drainages are provided to divert the surface and groundwaters from the road surface, which not only manages the torrential monsoon rains but also protect mountainous terrain from water-induced landslides.



b. Road aggregates are those materials that are used in road construction responsible for bearing stresses occurring on roads and also resist wear due to abrasive traffic action. It is necessary to carry out various tests on aggregates to ensure that undesirable materials are excluded from the highway pavements and best available aggregates are included.

Following test are carried out on the aggregates. They are:

- i. Descriptive tests
- ii. Nondestructive quality tests
- iii. Destructive tests
- iv. Specific gravity test

i. Descriptive tests:

- Descriptive tests are the visual examination of an aggregates that describes shape, surface texture of particles in the aggregate.
- The particle shape may be described as rounded, irregular, flaky, elongated, angular or both flaky and elongated.
- Surface texture may be defined as glossy, smooth, granular, rough, crystalline and honeycombed and porous.
- They are most useful in classifying aggregates.

ii. Nondestructive test:

This test is carried out on aggregate to determine its suitability for a specific use.

The results obtained are normally compared with aggregate specifications to see whether they comply with the desired properties and characteristics.

A. Gradation test:

- Tests is carried out by separating the aggregates into portions which are retained on a number of sieves or screens having specified openings which are suitably graded from coarse to fine.
- The result obtained may be expressed as total percentages retained between successive sieves.
- The total percentage passing method is very convenient for the graphical representation pf grading and most widely used in graded aggregate specifications.

B. Shape tests:

- The shape of aggregate mass is determined by the percentages of flaky, elongated and angularity of the particles.
- The evaluation of shape of the particles is made in terms of flakiness index, elongation index and angularity number.

C. Flakiness index:

- Flakiness index is the percentage by weight of the aggregate's particles whose least dimension or thickness is less than 0.6 times of their mean dimension.
- The test is applicable to aggregate sizes larger than 6.3 mm.

D. Elongation index:

- Elongation index of an aggregate is the percentage by weight of particles whose greatest dimension or length is greater than 1.8 times their mean dimension.
- The elongation test is not applicable for aggregate sizes smaller than 6.3 mm.
- It shouldn't exceed 15% for a good aggregate and normally shouldn't exceed 25%.

E. Angularity number:

- Angularity tests indicate the voids in excess of voids in perfectly rounded gravel
- More angular is the aggregate, the higher will be its angularity number.
- Angularity number of good aggregates should lie between 0 and 10.
- Angularity tests cannot be applied to the aggregates that get crushed during this test.

iii. Destructive tests:

A. Crushing test:

- It gives relative measure of the resistance of an aggregate to crushing under gradually applied compressive load.
- Aggregate with lower crushing value indicates a lower crushed fraction under load and giver longer service life to the road and hence more economical performance.
- Aggregate crushing value for surface course should be less than or equal to 30% and that of base course should be less than or equal to 45%.

B. Abrasion test:

- The test determines the resistance of aggregates to abrasion by Los Angeles Abrasion test.
- The principle of Loa Angeles abrasion test is t produce the abrasive action by use of standard steel balls mixed with aggregates and rotated drum for specific number of revolutions and cause impact on aggregates.
- For cement concrete abrasion value should be less than or equal to 16%.
- For bituminous mix surface course abrasion value is less than or equal to 30%.

C. Impact test:

- The property of material to resist the impact on it is known as toughness.
- Road aggregates are subjected to impact load due to movement of the vehicles. Hence, the aggregate should have sufficient toughness to resist their disintegration due to their impact.
- Aggregates with Aggregate Impact value less than or equal to 10% is exceptionally strong.

D. Soundness test:

- It determines the resistance of aggregates to weathering action by conducting accelerated weathering test cycle.
- The average loss in weight after 10 test cycles should not exceed 12% for the test with sodium sulphate or 18% for test with magnesium sulphate.

iv. Specific Gravity test:

- The test measures the quality or strength of material.
- Aggregates having low specific gravity values are generally weaker than those having higher values.
- Specific gravity of the road aggregates generally varies from 2.6 to 2.9.

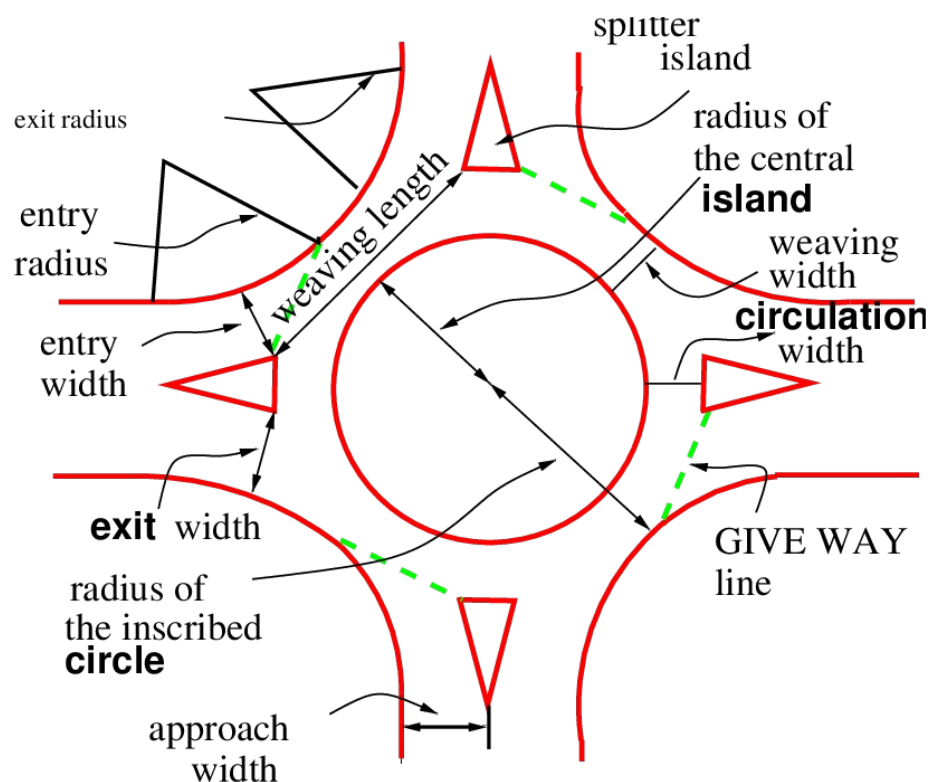
4. a. What is rotary intersection? What are the advantages and limitations of traffic rotary intersection? [5]

b. What are the main cause of traffic accidents in Kathmandu? What measures should we take to mitigate them. [5]

Answer:

a. Rotary intersection is an enlarged road intersection where all the converging vehicles are forced to move around a large central island in clockwise direction before they weave out of a traffic flow into their respective directions radiating from central island.

Rotaries and roundabouts are channelized intersections comprising of a central circle surrounded by one way roadway.



Rotary intersection is provided to eliminate unnecessary stopping for crossing stream of vehicles and reduce area of conflict.

Advantages:

- i. Crossing maneuvers are converted into weaving, merging and diverging operations.
- ii. The capacity of rotary intersection is the highest of all the other intersections.
- iii. Equal opportunity for all the vehicles as those turning left.
- iv. Operation cost at rotary intersection is less than that of signalized intersection where the vehicles have to stop and proceed.
- v. Rotary could function itself without the need of traffic police and signal.
- vi. Number of road crashes and its severity is reduced.
- vii. Rotaries can be constructed with advantage when the number of intersecting roads is between four and seven.

Limitations:

- i. Rotary intersection requires large area of land.
- ii. The operation in rotary intersection might be complex for the mixed traffic conditions.
- iii. If the pedestrian traffic is high in urban area, then rotary itself cannot control the traffic and requires traffic police.
- iv. When the traffic volume is very low, construction of a rotary cannot be justified.

b. Kathmandu is one of the most populated and congested cities in the world, with mixed traffic conditions. The roads are constructed smaller but the traffic density is increasing day by day. This has resulted in traffic congestion in increased duration and frequency and also contributed to increasing number of traffic accidents. According to traffic police statistics, about 12,000 people died in road accidents in the last 5 years. Out of this, about 1,000 people lost their lives in Kathmandu Valley. Last year alone, there were about 10,000 accidents in the valley. More than 200 people died in it.

The major causes of traffic accidents in Kathmandu are listed below:

- i. One of the major causes of traffic accident is over-speeding and reckless driving. Over-speeding on the narrow roads of Kathmandu designed for lesser speeds often causes collision resulting in damage to the vehicles as well as loss of life. As per police record, as many as 2,369 people lost their lives in road accidents in the previous fiscal year leaving hundreds of others injured. As per the police assessment, over-speeding is the main cause of accidents. The uncontrolled speeding led to over half of the accidents.
- ii. Inadequate driver training and licensing might also be another prominent cause of accidents in Kathmandu. Inexperienced drivers in the narrow road networks with mixed traffic system often lack the technique of proper driving maneuvers leading to accidents.
- iii. Drunk driving, distracted driving also contributes to road accidents. Consumption of intoxicants like alcohol might blur the vision as well as develop problems such as hallucinating. During such conditions it is difficult for a person to conduct driving operations consciously leading to road accidents. The drink and drive continue even as the traffic police under the Nepal Police have been carrying out an anti-drink and drive campaign for the last 15 years. Hundreds of cases of drink and drive are reported every month which is also a cause of significant numbers of accidents.
- iv. Jaywalking is another cause of traffic accidents in Kathmandu. Jaywalking and traffic rule violations has led to numerous accidents in Kathmandu Valley.
- v. Narrow and congested road, poor road surface conditions such as potholes and corrugations invite accidents.
- vi. Weak traffic law enforcement and improper parking are also another cause of road accidents.
- vii. Natural conditions such as rainfall, fog, snow etc. also reduce the visibility thus increasing the occurrence of road accidents.

Measures to mitigate traffic accidents:

- i. Road infrastructures must be improved by widening the narrow roads wherever possible, repairing the surface irregularities such as potholes and corrugations.
- ii. Proper Road sign, signals, road markings and traffic signals are installed.
- iii. Construction of pedestrian crossings, footpaths and overpasses at the required points.
- iv. Driver training and licensing procedures should be strengthened.
- v. Road safety awareness campaigns must be conducted regularly.
- vi. Reliable and efficient public transportation must be constructed.

5. **a. What are the functions of highway drainage? List the data necessary to glean before deciding on the drainage system for road.**

b. A bituminous mix has been prepared with 10% asphalt by weight of mixture. Assuming the specific gravity of asphalt to be 1 and that of void less specimen of the mixture to be 2.3, calculate the effective specific gravity of the aggregate.

Answer:

a. Highway drainage can be defined as the process of interception and removal of water from over, under and vicinity of road surface. It can be surface drainage or subsurface drainage. Removal and diversion of surface water from highway and adjoining land is called surface drainage. Similarly, removal of water from below the ground surface is called subsurface drainage.

The primary functions of a well-designed highway drainage system include:

- i. **Protection of Road Structure:**
It prevents the road surface and subgrade from softening and losing their structural strength and bearing capacity.
- ii. **Infrastructure Safety:**
By collecting and safely disposing of water, it protects pavements, embankments, and road users from water-related hazards.
- iii. **Erosion and Slope Control:**
It prevents the erosion of side slopes and protects mountainous terrain from water-induced landslides and the formation of cross ruts.
- iv. **Groundwater Management:**
It lowers the groundwater table (GWT) and prevents moisture accumulation beneath pavements, which could otherwise lead to frost action or soil instability.
- v. **Runoff Diversion:**
It prevents surface water from adjoining land from entering the roadway and ensures that water on the carriageway is drained off before it can percolate into the subgrade.

Before selecting or designing a drainage system for a road, several types of technical data must be gleaned to ensure the system's adequacy:

1. Hydrological Data:

- **Rainfall Characteristics:** Data on rainfall amount, duration, frequency, and intensity (IDF curves) is essential for estimating the maximum runoff the system must handle.
- **Catchment Area:** The size and extent of the area contributing runoff to the drainage system must be accurately measured.
- **Runoff Coefficient:** This value must be determined based on the surface types in the area (e.g., concrete, asphalt, or grass) as it affects how much water will flow into the drains.

2. Topographical and Design Data:

- **Terrain Classification:** Whether the land is plain, rolling, mountainous, or steep affects the velocity and volume of flow.

- **Road Geometry:** Information on longitudinal gradients, cross-slopes (camber), and the overall alignment of the road is necessary to plan flow paths.
- **Outfall Locations:** Identifying suitable natural drains or channels for the final disposal of collected water is vital.

3. Geotechnical and Environmental Data:

- **Soil Properties:** The soil type, classification, and bearing capacity must be known to design stable trenches and structures.
- **Soil Permeability:** This determines how quickly water moves through the ground and influences the design of subsurface drains.
- **Groundwater Table (GWT):** The highest expected level of the groundwater table must be identified to determine if subsurface drainage is required to keep the water at least 1.2 m below the subgrade.
- **Climatic Variations:** Beyond rainfall, data on local temperatures and potential for frost can influence the depth and type of drainage materials use.

b. **Given,**

Asphalt content by weight of mixture, $P_b = 10\% = 0.10$

Aggregate content by weight of mixture,

$$P_s = 1 - P_b = 1 - 0.10 = 0.90$$

Specific gravity of asphalt,

$$G_b = 1.0 \quad G_{sb} = 1.0$$

Specific gravity of voidless (maximum theoretical) bituminous mix,

$$G_{mm} = 2.3$$

$$\frac{1}{G_{mm}} = \frac{P_b}{G_b} + \frac{P_s}{G_{se}}$$

Substitute the given values:

$$\frac{1}{2.3} = \frac{0.10}{1} + \frac{0.90}{G_{se}}$$

$$0.4348 = 0.10 + \frac{0.90}{G_{se}}$$

$$\frac{0.90}{G_{se}} = 0.3348$$

$$G_{se} = 2.69$$

6. **A. What are the categories of maintenance as defined by DOR? What is the scope of periodic maintenance for paved road?**

b. How is the depth of scour determined for the design of bridge foundations? [5]

Answer:

a. Highway Road maintenance is the task of preserving, repairing and restoring a system of roadways to keep the serviceable conditions of highway as normal as possible and best as practicable. It is importance as

maintenance helps in providing better service facilities longer life and better appearance. It involves variety of planning, programming and scheduling to actual implementation in the field and monitoring.

Highway Maintenance activities can be classified into three categories as defined by DoR in DoR strategy (2016):

i. Routine maintenance activities:

- Those activities that are carried out as frequently as required during each year on all elements of the highway to improve its serviceability in all weather conditions are routine maintenance activities.
- Includes:
 - a. Grass and bush cutting, grading, reshaping of the unpaved surfaces.
 - b. Cleaning of carriageway, drains, slopes, culverts.
 - c. Replacement of structures that has been damaged.



ii. Recurrent maintenance activities:

- Those maintenance activities of localized nature carried out more or less at regular intervals of six months to 2 years are recurrent maintenance activities.
- Includes:
 - a. Sealing cracks, local surface treatment
 - b. Repair of depression and ruts
 - c. local constructions



iii. Periodic maintenance activities:

- Those activities done at the interval of several years to improve the overall functionality of road are periodic maintenance activities.
- Includes:
 - a. Renewal of wearing course of the pavement surface.
 - b. Restoration of road markings
 - c. Repainting of metal bridges etc.



iv. Special maintenance activities:

- Those activities of specialized and emergency nature done to solve special problems which may cause pavement failure if not identified on time.
- Includes:
 - a. construction of temporary diversion to allow the traffic to pass through obstruction.
 - b. Removal of debris and other obstacles.
 - c. Strengthening of pavement structures.

Road maintenance can also be grouped as:

i. Preventive maintenance:

It included actions that are taken to prevent premature deterioration and to retard the progression of deficiencies to reduce the rate of deterioration and increases useful life of pavement.

ii. Corrective maintenance:

It includes the action that includes actions taken to correct the deficiencies which are hazardous which seriously affect a pavement's operation to keep the highway within a tolerable level of serviceability.

DoR is also actively working on modernizing highway maintenance by introducing Nepal's first **Road Repair Ambulance (Infrastructure Ambulance)** for rapid repairs of pothole and a mobile reporting system for public complaints. These initiatives aim to provide faster maintenance, improve road safety, and enhance the overall efficiency of highway maintenance.

Scope of periodic maintenance for paved road:

Periodic maintenance covers all the long-term operations required within the service life of road and required only at interval of several years. It includes:

- i. Renewal of wearing course of pavement surface.
- ii. Resealing by surface treatment, slurry sealing
- iii. Resurfacing by construction of thin overlay
- iv. Re-gravelling of unpaved roads
- v. Surface dressing of paved roads
- vi. Restoration of road markings
- vii. Restoration of culverts and ancillary items
- viii. Repainting of metal bridge
- ix. leveling course to correct rutting, depressions, waves and undulations.
- x. Major drainage and structure repairs undertaken with the surfacing program

b. Scour depth is the measured depth of erosion around foundations caused by flowing water. Determination of scour depth is necessary to ensure that bridge foundations remain safe and stable during floods.

Depth of scour determined for the design of bridge foundations are as follows:

- i. Determination of design flood discharge:
Design flood of 50-year, 100 years etc. can be estimated depending upon bridge importance.

Design flood discharge is obtained using hydrological analysis.

- ii. Calculate normal scour depth:

For alluvial rivers, the normal scour depth is commonly estimated using Lacey's formula:

$$d = 0.473 \left(\frac{Q}{f} \right)^{1/3}$$

where :

d = normal scour depth (m)

Q = design discharge (m^3/s)

f = Lacey's silt factor

The silt factor is:

$$f = 1.76 \sqrt{m}$$

where :

m = mean particle size of bed material (mm)

- iii. Apply correction factors:

Adjust the normal scour depth for site conditions such as:

River bends (higher scour on the outer bend)
Bridge constrictions
Skew flow
Pier shape and spacing
Local turbulence

- iv. Determine maximum scour depth:

According to common bridge design practice (e.g., IRC recommendations for alluvial rivers):

- **Straight reach:** Maximum scour depth = $1.27 \times$ normal scour depth
- **Moderate bend:** Maximum scour depth = $1.50 \times$ normal scour depth
- **Severe bend or right-angle bend:** Maximum scour depth = $1.75 \times$ normal scour depth

- v. Fix the foundation level:

The foundation should be taken below the calculated maximum scour depth as specified by the relevant bridge design code, to ensure stability against scour and other forces.

7. a. Derive an expression for active earth pressure on the vertical back face of a E.W. supporting granular soil having horizontal surface. [6]

b. What is the effect of a) depth b) breadth c) ground water level on the bearing capacity of cohesive soils as per Terzaghi's formula. [1+1+2]

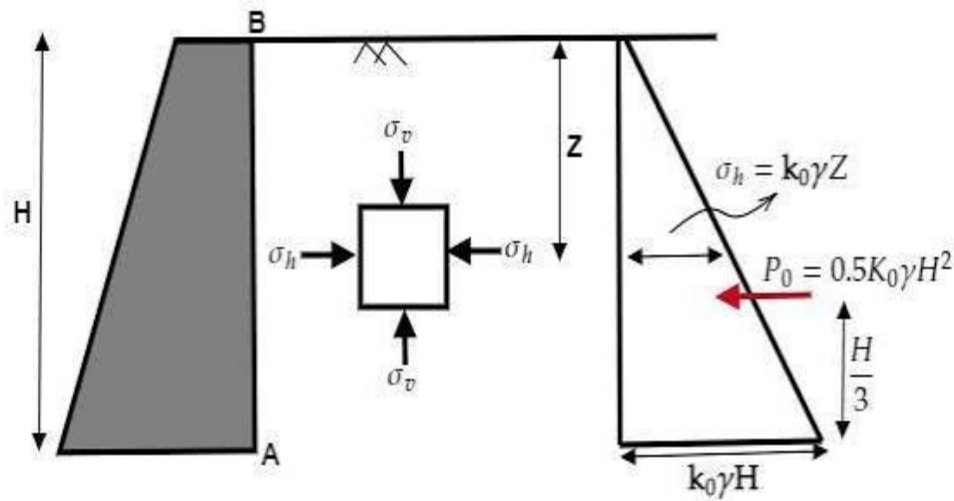
Answer:

The expression for active earth pressure on the vertical back face of earthwork supporting granular soil having horizontal surface is given by Rankine's theory.

The assumptions for Rankine's theory:

- i. The back of the retaining wall is vertical and smooth (no wall friction).
- ii. The backfill is dry, homogeneous, cohesionless (granular) soil.
- iii. The ground surface is horizontal.
- iv. The soil is in the plastic equilibrium state.

- v. Failure occurs along a plane surface.



Let us consider a soil element at a depth h below the ground surface.

Vertical stress due to overburden:

$$\sigma_v = \gamma h$$

where:

$$\gamma = \text{unit weight of soil (kN/m}^3\text{)}$$

$$h = \text{depth below ground level (m)}$$

The horizontal stress under active condition is

$$\sigma_h = K_a \sigma_v$$

where K_a is the coefficient of active earth pressure.

According to Rankine's failure criterion,

$$K_a = \frac{1 - \sin \phi}{1 + \sin \phi}$$

or

$$K_a = \tan^2 \left(45^\circ - \frac{\phi}{2} \right)$$

where

ϕ = angle of internal friction of soil.

Substituting

$$\sigma_h = K_a \gamma h$$

For a retaining wall of height H ,

Pressure at the top:

$$p=0$$

Pressure at the bottom:

$$p=K_a \gamma H$$

The total active earth pressure equals the area of the pressure triangle.

$$P_a = \frac{1}{2} \times \text{Base} \times \text{Height}$$

$$P_a = \frac{1}{2} (K_a \gamma H) (H)$$

Therefore,

$$P_a = \frac{1}{2} K_a \gamma H^2$$

This force acts horizontally on the wall.

Since the pressure distribution is triangular, the resultant acts at the centroid of the triangle.

Hence,

$$\text{Point of application} = \frac{H}{3} \text{ above the base}$$

b. For a strip footing on cohesive soil ($\phi=0$), Terzaghi's bearing capacity equation is:

$$q_u = c N_c + \gamma D_f N_q + \frac{1}{2} \gamma B N_\gamma$$

For cohesive soil ($\phi=0$):

$$N_c = 5.7$$

$$N_q = 1$$

$$N_\gamma = 0$$

Therefore,

$$q_u = 5.7 c + \gamma D_f$$

where:

q_u = ultimate bearing capacity

c = cohesion of soil

γ = unit weight of soil

D_f = depth of foundation

B = width (breadth) of footing

i. **Effect of the depth of foundation:**

↳ the equation:

$$q_u = 5.7 c + \gamma D_f$$

- Surcharge increases with the increase in depth of foundation.
- Hence, ultimate bearing capacity increases linearly with depth.

ii. **Effect of breadth of footing:**

For cohesive soil:

$$N_\gamma = 0$$

Hence, the term

$$\frac{1}{2} \gamma B N_\gamma = 0$$

- Breadth of footing has no direct effect on ultimate bearing capacity of cohesive soil according to Terzaghi.
- But, increase in the width of footing, reduces settlement and increases the total load the footing can carry.

iii. **Effect of Groundwater level:**

- Groundwater table affects the effective unit weight of the soil.
- Rise of GWT reduces the bearing capacity of foundation.

8. **a. Explain soil as three phase system with neat sketch, define the terms and give their interrelationship with usual notations: water content, degree of saturation, porosity, void ratio and unit weight of water, solid and soils at different states.**

Answer:

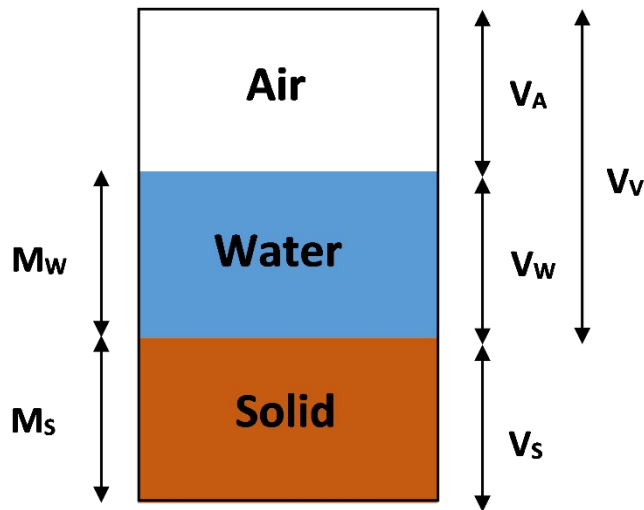
Soil is a three-phase system consisting of solid particles, water, and air. The voids between soil particles are occupied by water, air, or both. The total volume of soil is the sum of the volume of solids and voids, while the total weight is the sum of the weights of solids and water.

Important relations:

$$V = V_s + V_v$$

$$V_v = V_a + V_w$$

$$W = W_s + W_w$$



Water content:

Water content is the ratio of the weight of water present in the soil to the weight of dry soil solids.

Formula:

$$w = \frac{W_w}{W_s}$$

or

$$w(\%) = \frac{W_w}{W_s} \times 100$$

where :

W_w = Weight of water

W_s = Weight of soil solids

Degree of saturation:

Degree of saturation is the ratio of the volume of water to the volume of voids.

$$S = \frac{V_w}{V_v}$$

or

$$S(\%) = \frac{V_w}{V_v} \times 100$$

where :

V_w = Volume of water

V_v = Volume of voids

Range : $0 \leq S \leq 100\%$

Porosity (n)

Porosity is the ratio of the volume of voids to the total volume of soil.

$$n = \frac{V_v}{V}$$

or

$$n(\%) = \frac{V_v}{V} \times 100$$

where :

$$V_v = \text{Volume of voids}$$

$$V = \text{Total volume}$$

Void ratio (e):

Void ratio is the ratio of the volume of voids to the volume of solids.

$$e = \frac{V_v}{V_s}$$

where :

$$V_v = \text{Volume of voids}$$

$$V_s = \text{Volume of solids}$$

Unit weight:

Unit weight of water: Weight of water per unit volume

$$\gamma_w = \frac{W_w}{V_w}$$

$$\gamma_w = 9.81 \text{ kN / m}^3 \approx 10 \text{ kN / m}^3$$

Unit weight of solids: Weight of soil solids per unit volume of solids.

$$\gamma_s = \frac{W_s}{V_s}$$

Also ,

$$\gamma_s = G_s \gamma_w$$

where G_s = Specific gravity of soil solids.

Unit weight of soil at different state:

i. Bulk Unit weight: weight of moist soil per total volume.

$$\gamma = \frac{W_s + W_w}{V}$$

ii. Dry unit weight: Weight of dry soil solids per total volume.

$$\gamma_d = \frac{W_s}{V}$$

iii. Saturated Unit weight: Weight of fully saturated soil per total volume.

$$\gamma_{sat} = \frac{W_s + W_w}{V}$$

(when all voids are filled with water)

iv. Submerged Unit weight: Effective unit weight of saturated soil under water.

$$\gamma' = \gamma_{sat} - \gamma_w$$

Interrelationship among soil parameters:

i. Relation between void ratio and porosity:

$$n = \frac{e}{1 + e}$$

or

$$e = \frac{n}{1 - n}$$

ii. Relation between water content, degree of saturation and void ratio:

$$Se = w G_s$$

or

$$S = \frac{w G_s}{e}$$

where:

S = Degree of saturation

e = Void ratio

w = Water content

G_s = Specific gravity of solids

iii. Dry Unit weight:

$$\gamma_d = \frac{G_s \gamma_w}{1 + e}$$

iv. Bulk Unit weight:

$$\gamma = \frac{(G_s + Se) \gamma_w}{1 + e}$$

v. Saturated unit weight:

For saturated soil (S=1):

$$\gamma_{sat} = \frac{(G_s + e) \gamma_w}{1 + e}$$

vi. Submerged Unit Weight:

$$\gamma' = \gamma_{sat} - \gamma_w$$

9. write short notes on:

a. Mohr columnb theory: [3]

b. Primary and secondary consolidation: [3]

c. Causes of slope movement and failure: [4]

Answer:

a. Mohr Columnb theory:

Mohr Column theory of shear strength is also known as Mohr-Columb's failure criteria. Initially, Mohr stated that "the shear strength of soil is the function of normal stress (σ) acting on the failure plane.

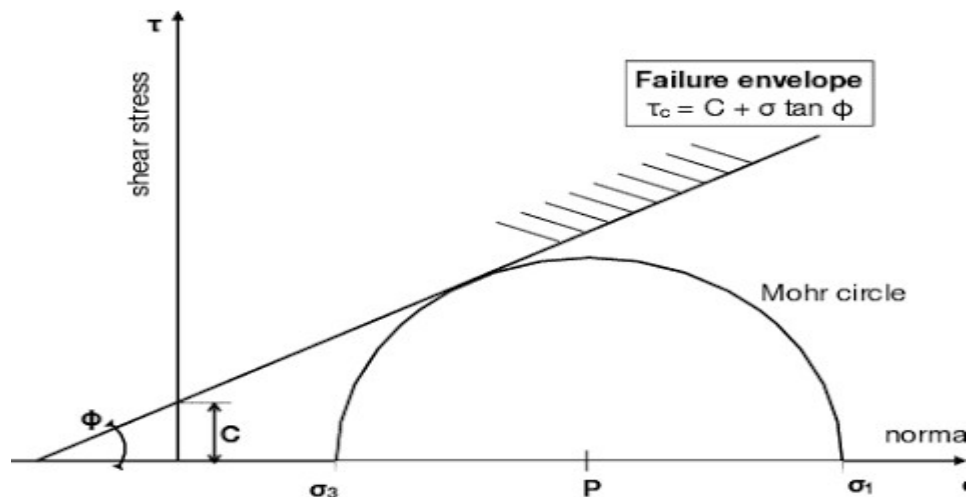
$$\text{i.e., } \tau = f(\sigma)$$

Later, Columb added on Mohr's theory stating "shear strength of soil is not only the function of normal stress (σ) but also depends on surface particles forces between soil particles."

$$\text{i.e., } \tau = c + \sigma \tan \phi$$

This theory is known as Mohr- Columb's theory of shear strength.

Mohr-Coulomb failure criterion is represented by a **straight-line failure envelope** on the shear stress (τ) versus normal stress (σ) graph.



Finally, Terzaghi modified Mohr-Columb's theory stating that "shear strength of soil is the function of effective stress not the total stress." This theory is known as modified Mohr-Columb's theory.

$$\text{i.e., } \tau = c' + \sigma' \tan \phi$$

b. Primary and secondary consolidation:**i. Primary consolidation:**

Consolidation that occurs when excess pore water is gradually expelled from soil voids is known as primary consolidation. It starts after the initial consolidation and is the most significant stages. As the water expelled out from the void, the settlement takes place which may take several days to year.

Settlement formula:

For normally consolidated clay:

$$S_p = \frac{C_c H}{1 + e_0} \log_{10} \left(\frac{\sigma'_0 + \Delta \sigma}{\sigma'_0} \right)$$

For over-consolidated clay:

If $\sigma'_0 + \Delta \sigma \leq \sigma'_c$:

$$S_p = \frac{C_r H}{1 + e_0} \log_{10} \left(\frac{\sigma'_0 + \Delta \sigma}{\sigma'_0} \right)$$

If $\sigma'_0 + \Delta \sigma > \sigma'_c$:

$$S_p = \frac{C_r H}{1 + e_0} \log_{10} \left(\frac{\sigma'_c}{\sigma'_0} \right) + \frac{C_c H}{1 + e_0} \log_{10} \left(\frac{\sigma'_0 + \Delta \sigma}{\sigma'_c} \right)$$

where:

S_p = Primary consolidation settlement

C_c = Compression index

C_r = Recompression index

H = Thickness of clay layer

e_0 = Initial void ratio

σ'_0 = Initial effective stress

$\Delta \sigma$ = Stress increment

ii. Secondary consolidation:

Consolidation in which occurs due to rearrangement of the soil particles after the completion of primary consolidation is known as secondary consolidation. It starts after all the excess pore water pressure has dissipated. The consolidation can continue for several years.

Formulae:

$$S_s = \frac{C_\alpha H}{1 + e_p} \log_{10} \left(\frac{t_2}{t_1} \right) S_s$$

where :

S_s = Secondary settlement

C_α = Secondary compression index

H = Thickness of compressible layer

e_p = Void ratio at the end of primary consolidation

t_1 = Time at end of primary consolidation

t_2 = Time after t_1

c. Causes of slope movement and failure:

If the shear strength of the soil is less than the shear stress developed, then the failure of slope takes place. The causes of slope failure can be categorized into two main groups:

i. Increase in shear stress on the slope:

Causes:

- a. Heavy raining causes seepage along the slope, seepage pressure acts downward and shear stress increases by seepage pressure.
- b. Increase in load on the slope after construction
- c. Increase in inclination of slope by erosion
- d. Dynamic loading: certain component acts downward.

ii. Decrease in shear strength of soil on slope:

Causes:

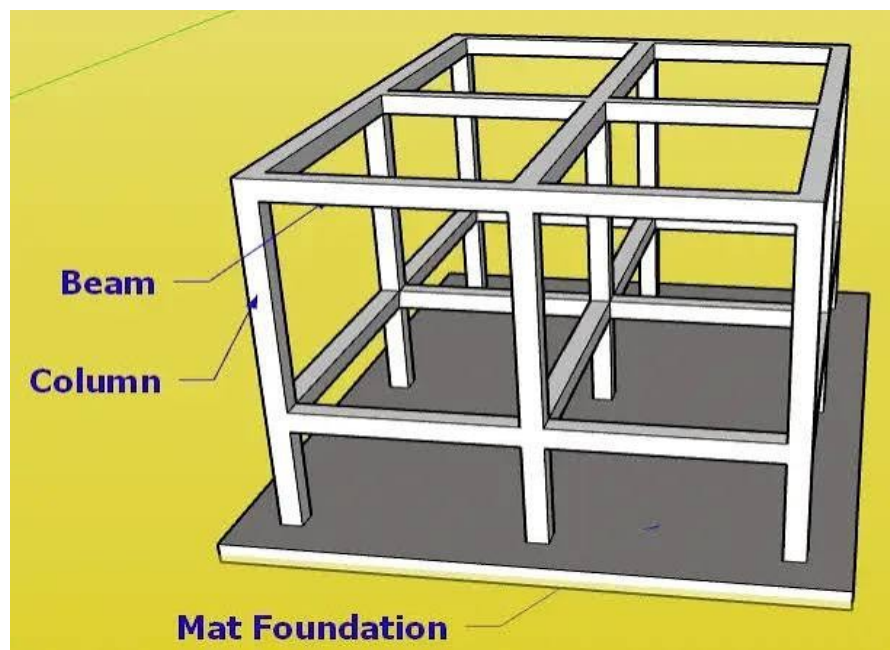
- a. Heavy raining increases the water on slope, increase pore water pressure and reduces effective stress and reduces the shear strength.
- b. Weathering and rock on slope
- c. Loss of fines and soil on slope by erosion.
- d. Cyclic dynamic loading

10. a) What are the foundation conditions under which mat foundation is preferred? 5

b) What is pile group efficiency? How is bearing capacity of a pile group determined in cohesive soils? 5 [5]

Answer:

- a. Mat foundation also known raft foundation is a large, continuous concrete slab that supports the weight of a building by spreading it across a large area.



The conditions under which mat foundation is preferred are as follows:

- i. Mat foundation is used when the soil has poor bearing capacity where it distributes the load over a larger area, reducing the contact pressure on the soil.'
- ii. When the columns are very close to each such that the individual footings overlap, then in that case mat foundation is more economical and practical.
- iii. For the structures subjected to heavy loads such as multistoried commercial buildings, mat foundations are used.
- iv. When there is possibility of equal settlements due to varying soil conditions, mat foundation is provided for uniform support and minimizes differential settlement.
- v. Mat foundation is also constructed where the soil is soft, compressible or has varying properties so that it spreads the load evenly on the ground.

- b. The ratio of the load-carrying capacity of a pile group to the sum of the individual capacities of all the piles in the group is known as pile group efficiency.

$$\eta_g = \frac{Q_g}{n Q_p}$$

Where :

η_g = Pile group efficiency

Q_g = Ultimate load capacity of the pile group

Q_p = Ultimate load capacity of one pile

n = Number of piles $\in$ the group

$\eta_g = 1$: Group capacity equals the $\sum$ of individual pile capacities.

$\eta_g < 1$: Negative group effect due to overlapping stress zones.

$\eta_g > 1$: Positive group effect (possible $\in$ dense granular soils due to compaction).

In cohesive soils (clay):

The ultimate bearing capacity of a pile group is determined by considering two possible modes of failure. Safe ultimate capacity is taken as the smaller of the two capacities.

i. Individual pile action:

The pile group acts as the sum of individual piles.

$$Q_{g1} = n Q_p$$

Where:

Q_{g1} = Capacity of pile group as individual piles

n = Number of piles

Q_p = Ultimate capacity of one pile

ii. Group action of piles:

When piles are closely spaced, the group and the enclosed soil behave as a single block.

The ultimate capacity is:

$$Q_{g2} = Q_b + Q_s$$

where ,

End Bearing Resistance: $Q_b = A_b N_c C_u$

For clay, $N_c = 9$

Hence, $Q_b = 9 C_u A_b$

Skin Friction Resistance: $Q_s = P L C_u$

$$Q_s = \alpha C_u P L$$

Where:

A_b = Area of the block base

P = Perimeter of the pile group block

L = Embedded length of piles

C_u = Undrained cohesion of clay

α = Adhesion factor (usually close to 1 for soft clays)

$$Q_{g2} = 9 C_u A_b + \alpha C_u P L$$

iii. Ultimate bearing capacity of piles:

The governing capacity is:

$$Q_g = \min(Q_{g1}, Q_{g2})$$

The allowable load is then: $Q_{\text{allowable}} = \frac{Q_g}{\text{Factor of Safety}}$ (Factor of Safety = 2.5 to 3).